

Project Views Queries

---1. Sales & Growth Performance section

-- First view: Total Sales & Profitability

-- Answer:

```
CREATE VIEW SALES_PERFORMANCE AS
SELECT
    SF.transaction_date,
    SF.product_id,
    SF.store_id,
    SF.customer_id,
    SF.quantity,
    PD.product_retail_price,
    PD.product_cost,
    (SF.quantity * PD.product_retail_price) AS Total_Revenue,
    (SF.quantity * PD.product_cost) AS Total_Cost,
    ((SF.quantity * PD.product_retail_price) - (SF.quantity * PD.product_cost)) AS
Total_Profit
FROM Sales_FACT AS SF
LEFT JOIN Products_DIM AS PD
ON SF.product_id = PD.product_id;
```

---1. Sales & Growth Performance section

-- Second view: Order Counts & AOV

-- Answer:

```
CREATE VIEW ORDER_ANALYSIS AS
SELECT
    SF.customer_id,
    SF.transaction_date,
    SF.store_id,
    SUM(PD.product_retail_price * SF.quantity) AS Order_Revenue,
    SUM(PD.product_cost * SF.quantity) AS Order_Cost,
    SUM(PD.product_retail_price * SF.quantity) - SUM(PD.product_cost * SF.quantity) AS
Order_profit
FROM Sales_FACT AS SF
LEFT JOIN Products_DIM AS PD
ON SF.product_id = PD.product_id
GROUP BY customer_id, transaction_date, store_id;
```

```
-- 2. Product & Quality Insights
-- First view: PRODUCT_SALES_ANALYSIS
-- Answer:
```

```
CREATE VIEW PRODUCT_SALES_ANALYSIS AS
SELECT
    SF.product_id,
    PD.product_brand,
    PD.product_name,
    SUM( SF.quantity ) AS Total_Qty_Sold,
    COUNT( SF.transaction_date ) AS Total_Sales_transactions
FROM    Sales_FACT    AS SF
LEFT JOIN Products_DIM AS PD
ON      SF.product_id=PD.product_id
GROUP BY SF.product_id,PD.product_brand,PD.product_name
```

```
-- 2. Product & Quality Insights
-- Second view: PRODUCT_RETURN_ANALYSIS
-- Answer:
```

```
CREATE VIEW PRODUCT_RETURN_ANALYSIS AS
SELECT
    RF.product_id,
    PD.product_brand,
    PD.product_name,
    SUM( RF.quantity ) AS Total_Qty_Returned,
    COUNT( RF.return_date ) AS Total_Return_transactions
FROM    Returns_FACT    AS RF
LEFT JOIN Products_DIM  AS PD
ON      RF.product_id=PD.product_id
GROUP BY RF.product_id,PD.product_brand,PD.product_name
```

```
-- 2. Product & Quality Insights
-- Third view: Brand performance
-- Answer:
```

```
CREATE VIEW BRAND_PERFORMANCE AS
SELECT
    PD.product_brand,
    PD.product_name,
    PD.product_sku,
    SUM(SF.quantity * PD.product_retail_price) AS Total_Revenue,
    SUM(SF.quantity * PD.product_cost) AS Total_Cost,
    SUM(SF.quantity * PD.product_retail_price) - SUM(SF.quantity * PD.product_cost) AS
Total_Profit,
    ((SUM(SF.quantity * PD.product_retail_price) - SUM(SF.quantity * PD.product_cost))
/ NULLIF(SUM(SF.quantity * PD.product_retail_price), 0)) * 100 AS Profit_Margin
FROM Sales_FACT    AS SF
JOIN Products_DIM  AS PD
ON SF.product_id = PD.product_id
GROUP BY PD.product_brand, PD.product_name, PD.product_sku;
```

-- 3. Customer segmentation
-- FIRST VIEW: Customer Income Analysis
-- Answer:

```
CREATE VIEW CUSTOMER_INCOME_ANALYSIS AS
SELECT    CD.yearly_income,
          SUM(SF.quantity) AS T_Prod_quantity,
          COUNT (DISTINCT CONCAT (SF.transaction_date,SF.customer_id )) AS TOTAL_TRANS,
          CAST ( SUM(SF.quantity)*1.0 / COUNT (DISTINCT CONCAT
(SF.transaction_date,SF.customer_id )) AS DECIMAL (10,2) )AS ITEMS_PER_ORDER
FROM      Sales_FACT AS SF
LEFT JOIN Customers_DIM AS CD
ON        SF.customer_id = CD.customer_id
GROUP BY CD.yearly_income
```

-- 3. Customer segmentation
-- Second VIEW: Customer Lifestyle Behavior
-- Answer:

```
CREATE VIEW CUSTOMER_LIFESTYLE_BEHAVIOR AS
SELECT    SF.customer_id,
          DM.homeowner,
          DM.num_children_at_home,
          COUNT( DISTINCT SF.transaction_date) AS T_TRANS
FROM      Sales_FACT AS SF
LEFT JOIN Customers_DIM AS DM
ON        SF.customer_id = DM.customer_id
GROUP BY SF.customer_id,
          DM.homeowner,
          DM.num_children_at_home
```

-- 4. Regional & Store Efficiency

-- VIEW: STORE Spatial EFFICIENCY

-- Answer:

```
CREATE VIEW STORE_SPATIAL_EFFICIENCY AS
SELECT      SD.store_id,
            SD.store_type,
            SD.total_sqft,
            SD.grocery_sqft,
            CAST( (SD.grocery_sqft*100.0) / SD.total_sqft AS decimal (10,2) ) AS
Utilization_Ratio,
            SUM(SF.quantity*PD.product_retail_price) AS STORE_TOT_SALES ,
            CAST (SUM(SF.quantity*PD.product_retail_price) / SD.total_sqft AS DECIMAL (10,2)
)AS SALES_PER_TOT_Sqft,
            CAST (SUM(SF.quantity*PD.product_retail_price) / SD.grocery_sqft AS DECIMAL
(10,2) )AS SALES_PER_GROCERY_Sqft
FROM        Sales_FACT AS SF
LEFT JOIN   Stores_DIM AS SD ON SF.store_id = SD.store_id
LEFT JOIN   Products_DIM AS PD ON SF.product_id = PD.product_id
GROUP BY   SD.store_id,SD.store_type,SD.total_sqft,SD.grocery_sqft
```

--5. Supply Chain Timing

-- VIEW: SUPPLY_CHAIN_EFFICIENCY

-- Answer:

```
CREATE VIEW SUPPLY_CHAIN_EFFICIENCY AS
SELECT      product_id,
            store_id,
            stock_date,
            transaction_date,
            DATEDIFF(day, stock_date, transaction_date) AS Days_On_Shelf,
            CAST(AVG(DATEDIFF(day, stock_date, transaction_date) * 1.0) OVER() AS
DECIMAL(10,2)) AS Global_Avg_Days_On_Shelf
FROM Sales_FACT;
```